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Fundamentals of Building AI Agents ✕

Today's goal • 0/3 items

Complete 3 items today to stay at a good pace!

Today's Skill Points ● 0 XP

See skill progress

Module 1

Foundations of Tool Calling and Chaining

Module 2

LCEL and Manual Tool Calling in LangChain

Introduction to Chaining and LCEL Basics

- ✓ LangChain LCEL Chaining Method
  Video • 4 min
- ✓ Lab: AI Powered Data Analysis with LCEL
  Ungraded App Item • 45 min
- ✓ Cheat Sheet: LangChain Expression Language (LCEL)
  Reading • 5 min
- ✓ Practice Quiz: Introduction to Chaining & LCEL Basics
  Practice Assignment • Grade: 100%

Manual Tool Calling Basics

- ✓ When to Call Tools Manually
  Video • 4 min
- ✓ Structured Outputs for Tool Calling
  Reading • 5 min
- ✓ Practice Quiz: Manual Tool Calling Basics
  Practice Assignment • Grade: 100%

Parsing and Validating Tool Calls

- ✓ Build LLM Agents with Tools
  Video • 4 min
- ✓ Build Interactive LLM Agents
  Video • 5 min
- ✓ Lab: Build Interactive LLM Agents with Tools
  Ungraded App Item • 1h
- ✓ Lab: Build a Tool Calling Agent
  Ungraded App Item • 1h
- ✓ Practice Quiz: Parsing and Validating Tool Calls
  Practice Assignment • Grade: 100%

Module Summary and Evaluation

- ○ **Summary and Highlights: Introduction to Chaining and LCEL Basics**
  Reading • 2 min
- ○ Cheat Sheet: Manual Tool Calling in LangChain
  Ungraded Plugin • 20 min
- ○ Graded Quiz: Manual Tool Calling in LangChain
  Graded Assignment • 21 min

Module 3

Using Built-in Agents in LangChain

Summary and Highlights: Introduction to Chaining and LCEL Basics

0:14 / 1:02 1x

Congratulations! You have completed this lesson. At this point in the course, you know:

- Large language models (LLMs) specify task parameters and suggest tools.
- Agents automatically invoke tools based on direction from the LLM.
- During manual invocation, developers verify inputs and outputs and adjust actions as needed.
- Manual invocation provides organizations with the opportunity for greater control, which enhances safety, manages costs, and can support enhanced accurate results.
- Mapping a dictionary connects tool names, such as `add`, to functions, enabling the large language model (LLM) to call the correct function by name and pass inputs as key-value pairs.
- You can manually control and validate tool inputs that the LLM can access by defining each tool using the `@tool` decorator.
- Before using `.invoke()`, bind the tools to the LLM so the LLM can identify and apply the correct tool based on the prompt.
- Use `.invoke(input_)` to pass a dictionary of key-value pairs that match the tool's parameter names.

Mark as completed

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